

Recommended Settings for Crash Stop

Electronic Equipment
BlueVision[®] | **NewGeneration**

E-INB-M-059-2016/00E



Power. Passion. Partnership.

Table of contents

1 SAFETY	3
1.1 GENERAL REQUIREMENTS	3
1.2 PERSONNEL AND ORGANIZATIONAL REQUIREMENTS	4
1.3 SAFETY REQUIREMENTS FOR COMMISSIONING AND OPERATION	5
1.4 CONVENTIONS FOR WARNINGS USED IN THE TEXT	6
2 FUNCTIONAL DESCRIPTION.....	7
2.1 CRASH STOP	7
2.2 SHIP MODEL.....	8
2.3 COUPLING SEQUENCE	9
2.4 SPEED OFFSET	11
3 RECOMMENDED SETTINGS FOR THE CRASH STOP FUNCTIONALITY.....	12
3.1 DEFAULT SETTINGS	12
3.1.1 <i>Checking gearbox engagement speed/window</i>	12
3.1.2 <i>Checking gearbox disengagement speed/window</i>	12
3.1.3 <i>Engagement test (HAT)</i>	13
3.1.4 <i>Speed offset</i>	13
3.2 PREREQUISITES.....	13
3.3 SETTING THE MAXIMUM SPEED WHEN ENGAGED ASTERN	13
3.4 SETTING THE SHIP MODEL	14
3.4.1 <i>Shaft speed and ship speed (pShipoverShaftSpeed_X0)</i>	14
3.4.2 <i>Time constants for acceleration performance</i>	16
3.4.3 <i>Time constants for deceleration performance</i>	18
3.5 SETTING PARAMETERS FOR THE “REVERSING” MANEUVER	20
3.5.1 <i>Preparatory steps</i>	20
3.5.2 <i>Establishing the max coupling limit (pNormalEngageShipSpeed)</i>	21
3.5.3 <i>Detecting the “reversing” coupling maneuver</i>	23
3.5.4 <i>Speed offset for coupling maneuver “reversing”</i>	23
3.6 CHECKING/SEQUENCE OF THE “REVERSING” COUPLING MANEUVER.....	24
3.7 SETTING PARAMETERS FOR THE “CRASH STOP” MANEUVER.....	25
3.7.1 <i>Establishing the max coupling limit (pCrashEngageShipSpeed)</i>	25
3.7.2 <i>Detecting a “crash stop” coupling maneuver</i>	26
3.7.3 <i>Speed offset for coupling maneuver “crash stop”</i>	26
3.8 CHECKING/SEQUENCE OF THE “CRASH STOP” COUPLING MANEUVER	27
4 CRASH STOP MODES.....	28
5 QUALIFICATION	29
6 CLOSING REMARKS.....	29
7 ANNEX	30
7.1 SHIP MODEL TABLE (PSHIPOVERSHAFTSPEED_X0)	30
7.2 PARAMETER OVERVIEW (REVERSING/CRASH STOP).....	30

1 Safety

1.1 General requirements

General information

This document specifies recommended settings for adjusting the responses of the BlueVision | NewGeneration automation system when a crash stop is initiated (SPU software version Var1_Ed01_040 and higher). The document is not intended as an alternative to formal training in initial start-up procedures and servicing at the Training Center.

The shipyard must be made aware of the absolute necessity of conducting a crash stop maneuver during trials in order to verify that the necessary settings have been made to ensure that the vessel responds safely and reliably in the event of an emergency.

Make a written entry in the commissioning record stating reasons, and have the shipyard confirm accordingly, if it is not possible to configure the crash stop function in the course of sea trials, or should the shipyard or customer fail to facilitate this.

Observe all generally applicable, national, statutory and otherwise relevant accident prevention and environmental protection regulations in addition to the instructions provided in this publication. This engine has been manufactured according to the current state of the art and applicable specifications and regulations. Nevertheless, the engine may endanger personnel or incur damage if:

- It is not put to its intended use
- It is not operated, maintained and repaired by trained personnel
- It is changed or modified
- Safety notes are disregarded

Intended use

The engine is intended for use in accordance with contractual agreements and/or the purpose for which it was designated on delivery. Any other use above or beyond this shall be deemed improper. The engine manufacturer accepts no liability whatsoever for damage incurred by improper use.

The user shall bear all risk in such case.

Intended use includes compliance with the operating instructions and strict observance of maintenance and repair procedures.

Changes or modifications

Unauthorized changes to the engine compromise safety.

MTU accepts no liability whatsoever for damage resulting from unauthorized changes or modifications and all warranties will be rendered null and void.

Spare parts

Use genuine MTU spare parts only when replacing components or assemblies. MTU accepts no liability whatsoever for damage resulting from the use of non-MTU spare parts and all warranties will be rendered null and void.

Corrective work on components

Work shall be entrusted to an authorized MTU workshop in case of repair or engine overhaul.

1.2 Personnel and organizational requirements

Personnel requirements

Work on the product shall be performed by trained and qualified personnel only:

- Trained at the manufacturer's Training Center
- Qualified in machinery and plant engineering

The user shall define personnel responsible for operation, maintenance, repair and transportation.

Organizational measures

This publication shall be made available to all personnel involved in operation, maintenance, repair or transportation.

It must be kept ready to hand in the vicinity of the engine and be readily available to operating, maintenance, repair and transportation personnel at all times.

This publication serves as a basis for instructing personnel in matters of handling and repair of the engine. Safety instructions, in particular, must be read and clearly understood.

This applies especially to personnel only working occasionally on the engine. Such personnel must receive repeated instruction to refresh their knowledge.

Use the Spare Parts Catalog to identify spare parts for maintenance and repair work.

Workwear and protective equipment

Wear safety clothing at all times when working on or around the engine.

Wear suitable personal safety equipment appropriate to the type of work concerned.

1.3 Safety requirements for commissioning and operation

Safety requirements for commissioning

The assembly/system must have been properly installed and accepted in accordance with MTU specifications before being put into operation for the first time.

Before starting up any device or system, ensure that:

- All maintenance and repair work has been completed.
- All loose parts have been removed from rotating parts of the machine.
- All personnel is clear of the danger zone around moving parts of the machine.

Immediately after starting up the device or the system ensure that all controls and display instruments including monitoring, signaling and alarm systems are operating correctly.

Safety requirements for operation

Emergency procedures must be practiced on a regular basis.

Operators must be familiar with the controls and displays.

Operators must be aware of the effects of any action they perform.

During operation, the indicators, displays and monitoring assemblies must be supervised with respect to current operating states, violation of limit values as well as warnings and alarm messages.

If a fault in the system is detected or signaled by the system:

- Inform the person in charge.
- Evaluate the signal or message.
- Take any necessary counteraction, e.g. emergency engine stop.

Engine operation

The following requirements must be fulfilled prior to engine starting:

- Wear ear protectors.
- Ensure that the engine room is well-ventilated.
- Do not inhale exhaust gases emitted by the engine.
- Check exhaust pipework for leakage and ensure that exhaust gases are vented to atmosphere.
- Wipe up any leaked or spilled fluids and lubricants or use a suitable binding agent.
- Protect battery and alternator terminals or cabling from inadvertent contact.
- Never undo water, oil, fuel, compressed air or hydraulic lines when the engine is running.

Electrical equipment

Some of the components are live when electrical equipment is in operation.

Observe warnings and instructions which apply to such equipment.

1.4 Conventions for warnings used in the text

DANGER 	In case of imminent threat or danger. Consequences: Death, severe or irreversible injury! • Corrective action.
WARNING 	In case of possible threat or danger. Consequences: Death, severe or irreversible injury! • Corrective action.
CAUTION 	In case of possible threat or danger. Consequences: Minor injury! • Corrective action.
ATTENTION 	In case of possible damage to the product. Consequences: Material damage! • Corrective action. • Additional information about the product.

Warnings

Personnel charged with operation, maintenance, repair or transportation must have read and understood this manual including all the safety instructions and warnings provided therein.

2 Functional description

2.1 Crash stop

Crash stopping involves bringing the vessel to a standstill as quickly as possible in the event of an emergency while simultaneously maintaining maneuverability (avoiding stalling the engine(s)).

Engine speed was the only criterion used to initiate the crash stop coupling sequence for BlueLine and SmartLine systems. Other influences such as current, wind and loading could not be taken into account. However, as these factors have a significant effect on the vessel's response to a crash stop, ship speed has now also been included as an additional criterion. This ensures that a coupling maneuver is always possible at a certain ship speed regardless of any external influences (e.g. wind, current, loading).

The system calculates ship speed on the basis of a theoretical model comprising engine speed and shaft speed.

The coupling sequence can be optimized by calculating ship speed in this way.

ATTENTION



The chapter 3 of this document describes the Extended Mode setup procedure.

The chapter 4 of this document describes the 2 additional modes „simplified Mode“ and „ShaftSpeed only mode“.

2.2 Ship model

Functional description:

The ship model is an RCS functionality. It is used to calculate the speed of the ship.

Ship speed is calculated using the shaft speed sensor(s).

The mean value of all shaft speed data is used to calculate ship speed on multiple shaft plants (e.g. twin-shaft propulsion arrangement).

In the absence of a shaft speed sensor, calculation is based on engine speed and the transmission ratio of the gearbox.

The ship model also takes the acceleration and deceleration characteristics of the ship into account.

Basic data:

Basic data must be determined in the course of sea trials and entered in the relevant parameters of the ship model.

Shaft speed and ship speed

The ship model must be completed with a supply of data on shaft speeds and the corresponding ship speeds. This is the static ship model.

The range between full astern and full ahead is represented by 10 grid points.

Time constants for acceleration and deceleration characteristics

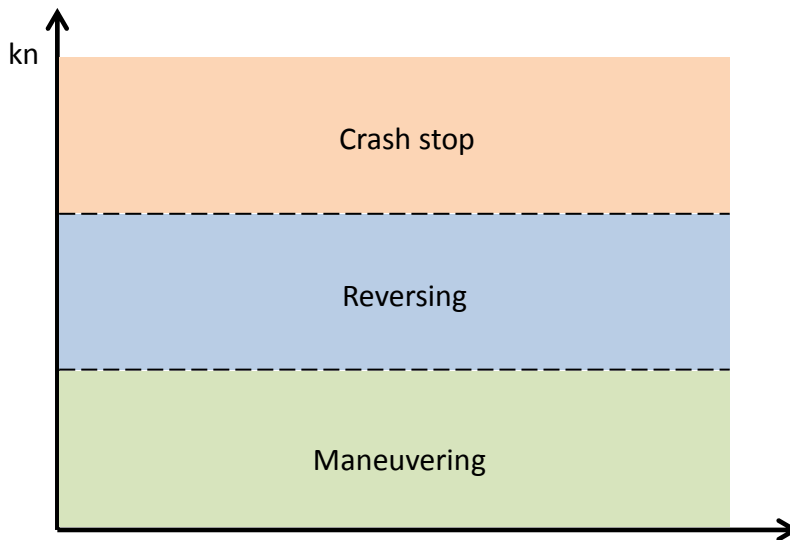
Time constants influence the dynamic acceleration and deceleration characteristics.

These values must be adapted accordingly depending on the characteristics of the static ship model compared with the actual response of the ship.

The static ship model has to be adapted as the various vessel types (planing craft, semi-planing craft, displacer hull) respond differently.

2.3 Coupling sequence

The coupling sequence of the BlueVision | NewGeneration automation system can be divided into three ranges.



In the first range, coupling (maneuvering) is possible at low speeds without significant delays (apart from those usually associated with this technology).

The clutch can be engaged and disengaged in this range without any additional measures (speed increase).

Note:

Should it already become apparent that an increase in speed is necessary at this point, it can be safely assumed that it will not be possible to achieve satisfactory engagement/disengagement and coupling at higher speeds by merely adjusting the settings in the BVNG system.

The customer and the shipyard must be consulted in such cases to clarify whether other measures can be adopted (e.g. by the gearbox manufacturer).

The second range begins at a defined ship speed at which coupling should also be possible without additional measures (speed increase) or significant speed dip and which also allows the range to be detected at the same time.

A coupling sequence initiated at a ship speed in this range is referred to as a “reversing” maneuver below.

The “reversing” range ends on detecting the third or “crash stop” range.

The third range begins on detecting the “crash stop” range.

Owing to the high speed of the ship in this range, coupling is no longer possible without stalling the engine unless additional measures (speed increase) are implemented.

By taking additional measures (increasing the speed) coupling is possible from higher ship speeds at an earlier point in time than is the case for the “reversing” sequence.

This range is referred to as a “crash stop” maneuver below.

The “reversing” and “crash stop” ranges are characterized by their own parameters for range detection, a definition of specific coupling limits and in case of “crash stop” an additional speed offset parameter.

The difference between coupling in the “maneuver” range and the other two ranges (“reversing” and “crash stop”) is that ship speed is also taken into account in the case of the latter in addition to the disengagement limit and the engagement window which depend on engine speed (to protect the gearbox).

A differentiation is made here between ship speed for disengagement and ship speed for engagement astern (reverse coupling) in the case of these two coupling sequences.

Ship speed for disengagement

The deceleration effect could be improved by leaving the gearbox engaged at the start of the coupling sequence until the propeller/engine no longer act as a brake.

This point in time can be established by monitoring the injection quantity which increases again even though engine speed continues to drop. However, this enhancement may be disregarded as it has little influence on the deceleration characteristics.

The ship speed limit for disengagement ***pShipSpeedDisengage*** applies to the “reversing” and “crash stop” coupling sequences and is usually set to 100kt by default.

As no vessel is likely to reach such a speed, this limit is not relevant for any subsequent steps. It shall therefore not be given further consideration in this document.

Background information:

As the ship speed limit for disengagement lies above the max. attainable ship speed, the speed demand and thus the engine speed is set directly to the increased speed value (engine idling speed + speed offset) in the event of a “crash stop” maneuver, rather than being reduced to idling before being increased by the speed offset once more.

This ensures that the speed offset is in effect when coupling takes place, i.e. the engine is running at the increased speed (engine idling speed + speed offset) when coupling is initiated.

The disengagement criterion is thus merely the maximum engine speed for disengagement specified by the gearbox manufacturer.

Ship speed for engagement astern

The ship speed at which engagement astern takes place is defined by separate coupling limits for the “reversing” and “crash stop” maneuvers.

Engagement astern can therefore take place at different ship speeds.

This allows the coupling sequence to be further optimized.

The following applies to the “reversing” and “crash stop” coupling maneuvers:

The gearbox is engaged astern when the following two criteria are fulfilled:

- 1.) Engagement window
Engine speed is below the maximum engagement speed
(maximum admissible engine speed for engagement specified by the gearbox manufacturer)
- 2.) Ship speed (coupling limit)
Ship speed is at or below the established rate at which engagement astern is possible without stalling the engine.

In summary:

“Maneuvering”, “reversing” and “crash stop” are all coupling sequences.

The “reversing” and “crash stop” maneuvers each have their own parameters which determine:

1. When ship speed is in the relevant range (range detection)
2. The ship speed at which coupling takes place (max. coupling limit)
3. How high the engagement speed is on coupling for the maneuver concerned (speed offset)

Basically two crash stop parameter settings (reversing/crash stop) are available for use.

2.4 Speed offset

The idling speed can be increased by an offset to assist the engine during the coupling sequence and prevent it from stalling or stopping.

Higher engine speed increases the injection quantity / engine power on coupling and helps counteract speed dip.

Also here there are for "normal clutching" while maneuvering, for maneuver "Reversing" as well as for maneuver "Crash Stop" suitable parameters which add up themselves.

3 Recommended settings for the crash stop functionality

Estimated time required:

1-2 hours

Note:

- ✓ **Established values must be implemented/entered for all shafts.**
- ✓ **All maneuvers shall be performed in SCL mode.**
- ✓ **Initiating a coupling sequence involves shifting the propulsion control lever from the ahead range directly to the astern range.
The propulsion control lever must not remain at neutral for period of time x in this case.**

3.1 Default settings

Check the clutch control parameters, adapt as necessary and test them in the course of harbor trials (HAT) before attempting to make the crash stop settings.

3.1.1 Checking gearbox engagement speed/window

One engagement window parameter is provided for Local mode and Remote mode respectively. Normally, these only need to be checked, but they may have to be adapted in some cases.

The values set must not exceed the limit values specified by the gearbox manufacturer to avoid damaging the gearbox and the clutch.

For Local mode the MCS parameter

pEngageWindowHiLimit (e.g. SysID 1.001.0020.043108.154.0000.3) must be used

and in Remote mode the RCS parameter

pEngageWindowHiLimit (e.g. SysID 1.002.0020.043108.154.0000.3).

3.1.2 Checking gearbox disengagement speed/window

One disengagement window parameter is also provided for Local mode and Remote mode respectively.

Normally, these only need to be checked, but they may have to be adapted in some cases.

The values set must not exceed the limits specified by the gearbox manufacturer.

For Local mode the MCS parameter

pDisengageWindowHiLimit (e.g. SysID 1.001.0020.043107.154.0000.3) must be used

and in Remote mode the RCS parameter

pDisengageWindowHiLimit (e.g. SysID 1.002.0020.043107.154.0000.3).

3.1.3 Engagement test (HAT)

Ensure that there is no excessive dip in engine speed on engaging normally in the course of harbor trials (HAT). Should this be the case, consult the customer and set an idle speed offset as necessary.

3.1.4 Speed offset

In case of excessive speed dip during normal engagement, a permanent idle speed offset can be set using the parameter ***pIdleSpeedOffset*** (SysID 1.00x.0010.000518.012.0000.3).

This step must be discussed with the customer as a permanent idle speed offset leads to increased wear-and-tear on the clutch and gearbox and also raises the speed when the propulsion control lever is set to ahead.

By using the parameter ***pEngageSpeedOffset*** (SysID 1.00x.0080.000517.012.0000.3) the engine speed will be increased during the coupling maneuver only (Maneuvering, Reversing, Crash Stop).

3.2 Prerequisites

The following prerequisites must be fulfilled to determine the settings for the crash stop functionality:

- ✓ Ship at sea (not in shallow water)
- ✓ Maximum ship speed possible in steady state
- ✓ Typical loading (Note: This affects planing craft differently to displacer hulls)
- ✓ Ship speed can be determined by on-board GPS receiver

3.3 Setting the maximum speed when engaged astern

The maximum speed admissible for the ship when traveling astern must be established in the course of trials (propulsion control lever set to full astern). This should be defined by the shipyard or the captain.

The default value is set to 67% of maximum engine speed and must be adapted as necessary using the configurator in the Automation Suite.

3.4 Setting the ship model

The speed of the vessel at a given shaft/engine speed must first be established in order to set the ship model. The entire range between full astern and full ahead is considered initially. Ten grid points within this range are entered in the parameters **pShipoverShaftSpeed_X0**. 5 of these grid points are preset, 5 other suitable grid points must be defined.

Acceleration and deceleration characteristics must also be determined.

3.4.1 Shaft speed and ship speed (pShipoverShaftSpeed_X0)

Determination of the vessel speed as a function of the shaft speed (calculated by the engine speed) using the Excel table

Note:

When opening the Excel table, the gear ratio is queried and must be entered. This automatically converts the engine speed into a shaft speed in the table.

1. Run the engine speed up from engaged ahead to full ahead in steps of 100 rpm.
Wait until a steady-state has been established then enter the following values in the Excel table (Annex):
 - ☒ Engine Speed **EngineSpeed** (SysID 1.00x.0010.000501.010.0000.0) and
 - ☒ Ship speed (read-off at the on-board GPS receiver)
2. Set the propulsion control lever to engaged astern and subsequently to full astern.
Wait until a steady-state has been established then enter the following values in the Excel table (Annex):
 - ☒ Engine Speed **EngineSpeed** (SysID 1.00x.0010.000501.010.0000.0) and
 - ☒ Ship speed (read-off at the on-board GPS receiver)

Note:

The maximum 24 points (values) are shown as a green characteristic in the graphic.
If not all 24 points are required, the values of full ahead must be used for the remaining points.

Creating the graphic representation and determining the grid points

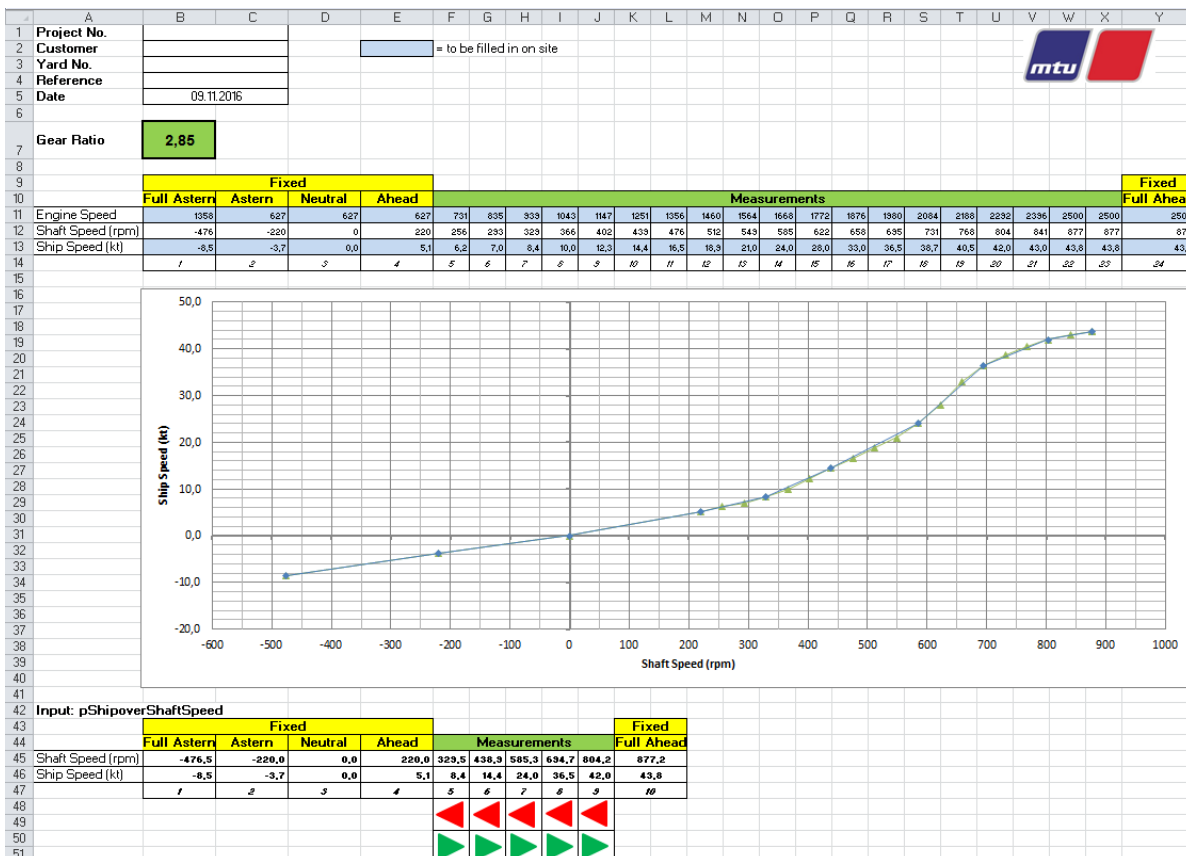
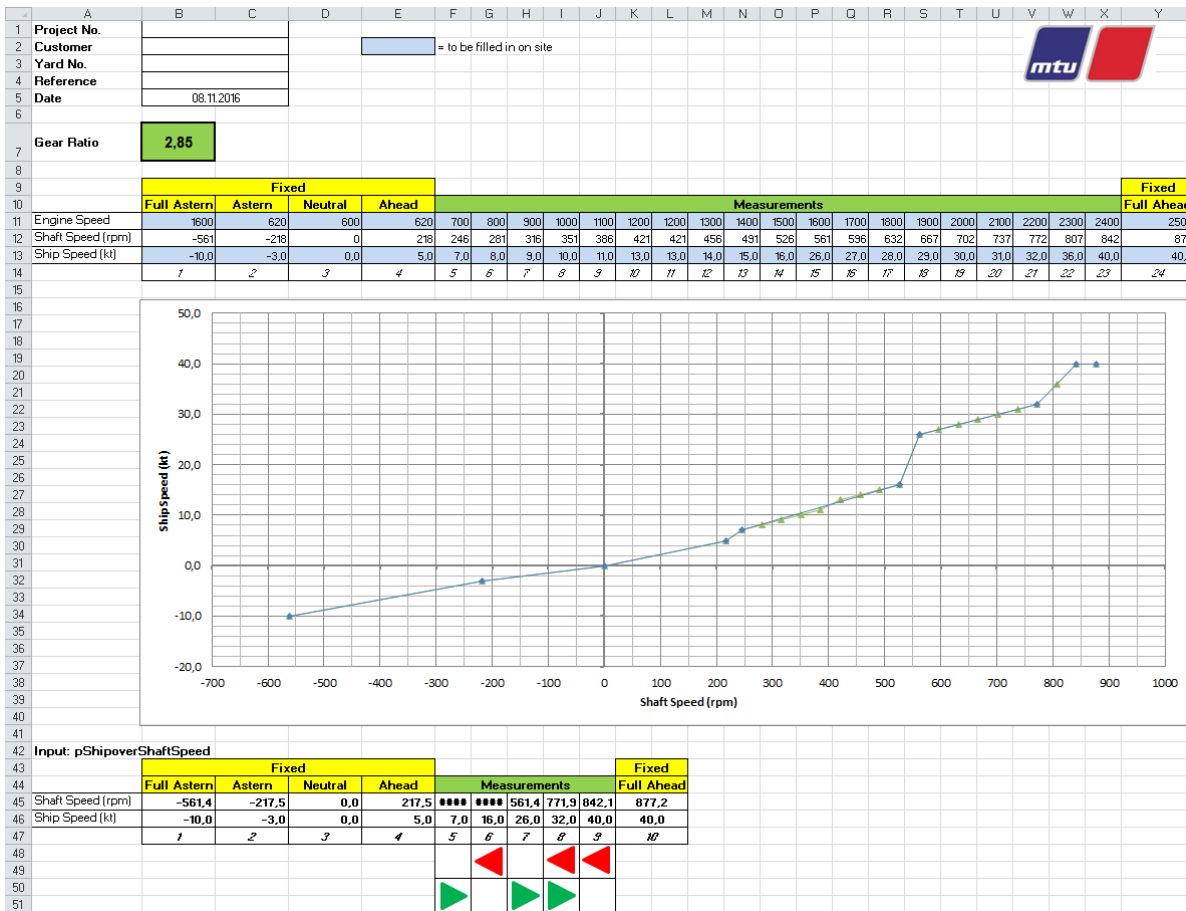
Note:

The 10 points (values for **pShipoverShaftSpeed_X0**) to be identified for the ship model are shown as a blue characteristic in the graph.

3. Always use the predefined stages (full astern, astern engagement lock, neutral engagement lock, ahead engagement lock and full ahead).
This leaves another 5 points between the ahead engagement lock and full ahead.
4. Use the green and red arrows (lines 48-51 in the Excel table) to move the 5 variable points until the blue characteristic covers or reflects the green characteristic.

Transferring grid points to ship model parameters

5. The 10 determined interpolation points (shaft speed, ship speed) must now be transferred to the corresponding 2D curve of the ship model **pShipoverShaftSpeed_X0** (SysID 1.00x.0050.000800.302.0001.3).



Example of selected grid points between engaged ahead and full ahead

3.4.2 Time constants for acceleration performance

The static ship model is adapted to the actual dynamic characteristics of the vessel in terms of acceleration performance with the time constant

pTimeConstant_ShipAcceleration (SysID 1.00x.0050.000804.012.0000.3).

This is determined by performing an acceleration maneuver.

The time the ship model requires to reach 90 % of the maximum ship speed is then compared with the time actually needed by the vessel.

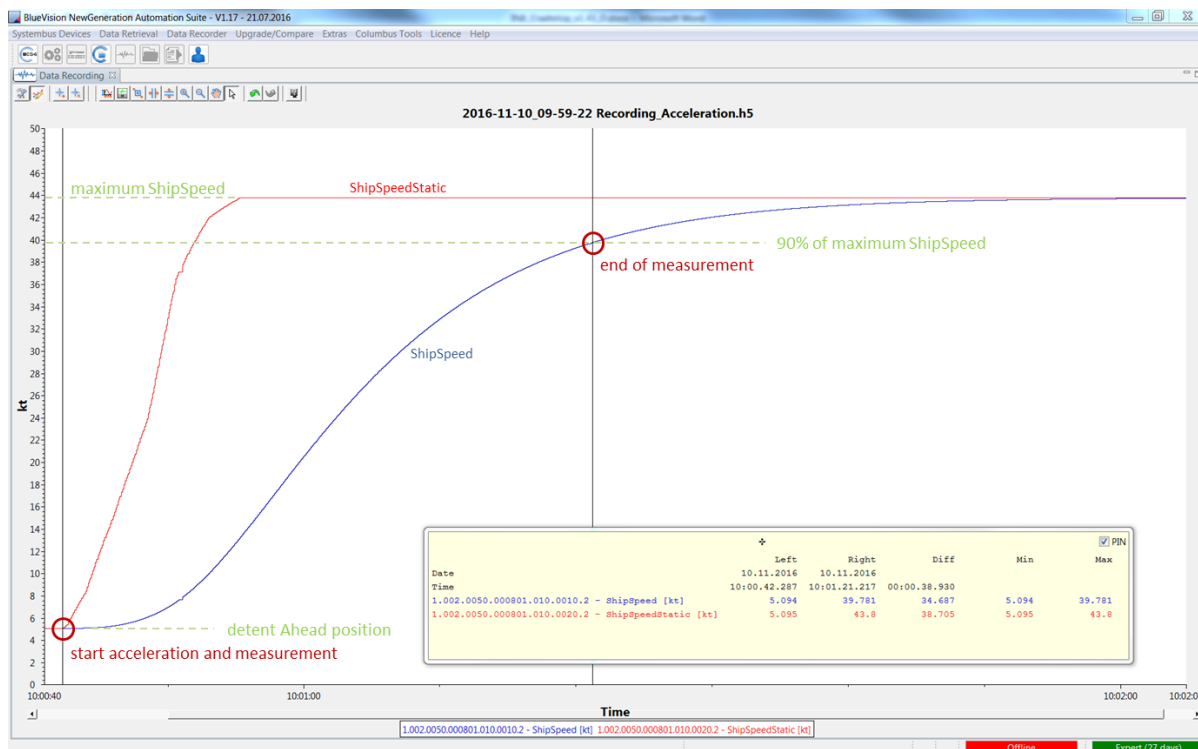
The ship model can then be adapted by means of the time constant in case of deviation.

Values to be established:

- ☒ Static ship speed of the ship model
ShipSpeedStatic (SysID 1.00x.0050.000801.010.0020.2)
- ☒ Calculated dynamic ship speed of the ship model
ShipSpeed (SysID 1.00x.0050.000801.010.0010.2)
- ☒ Time required by the ship to reach 90% of the maximum speed (measured with a stop watch)
- ☒ Actual ship speed (read-off at the on-board GPS receiver)

Proceed as follows to establish the time constant for acceleration performance:

1. Start recording the measuring points of the dynamic ship model **ShipSpeed** (SysID 1.00x.0050.000801.010.0010.2) and the static ship model **ShipSpeedStatic** (SysID 1.00x.0050.000801.010.0020.2).
2. Set the propulsion control lever to the engaged ahead lock in SCL mode.
Wait until the corresponding speed has been reached.
3. Move the propulsion control lever from the engaged ahead lock directly to the full ahead setting.
4. Start recording the time with a stop watch at this point.
5. Stop time measurement at the stopwatch at 90% of the maximum real vessel speed (read-off at the on-board GPS receiver) and record the time.
6. Stop recording after reaching the maximum speed.
7. Evaluate the recording after completing the maneuver as described below.



Timing starts with the left cursor position at the start of the acceleration maneuver and ends with the right cursor position at 90% of the maximum speed.

This period equates to the acceleration time required by the ship model.

Subsequently compare the two times thus measured (actual time measured with the stop watch and the time required by the ship model).

If the time the ship model needs to attain maximum speed is greater than the time the vessel actually needs to accelerate, the parameter ***pTimeConstant_ShipAcceleration*** (SysID 1.00x.0050.000804.012.0000.3) must be reduced (ship model accelerates faster).

If the time the ship model needs to attain maximum speed is less than the time the vessel actually needs to accelerate, the parameter ***pTimeConstant_ShipAcceleration*** (SysID 1.00x.0050.000804.012.0000.3) must be increased (ship model accelerates slower).

3.4.3 Time constants for deceleration performance

The static ship model is adapted to the actual dynamic characteristics of the vessel in terms of deceleration performance with the time constant ***pTimeConstant_ShipDeceleration*** (SysID 1.00x.0050.000804.012.0010.3).

This is determined by performing a deceleration maneuver.

The time taken by the ship model to decelerate from the maximum vessel speed to the speed entered in the Excel table (for the ship model) at 1100 RPM is compared with the time the ship actually needs.

The ship model can then be adapted by means of the time constant in case of deviation.

Values to be established:

- ☒ Static ship speed of the ship model
ShipSpeedStatic (SysID 1.00x.0050.000801.010.0020.2)
- ☒ Calculated dynamic ship speed of the ship model
ShipSpeed (SysID 1.00x.0050.000801.010.0010.2)
- ☒ Time required by the ship to reach the speed entered in the Excel table at 1100 RPM (measured with a stop watch)
- ☒ Actual ship speed (read-off at the on-board GPS receiver)

Proceed as follows to establish the time constant for deceleration performance:

1. Accelerate the ship to maximum speed.
Wait until the ship has reached its maximum speed.
2. Start recording the measuring points of the dynamic ship model ***ShipSpeed*** (SysID 1.00x.0050.000801.010.0010.2) and the static ship model ***ShipSpeedStatic*** (SysID 1.00x.0050.000801.010.0020.2).
3. Move the propulsion control lever in SCL mode from the full ahead setting directly to the neutral engagement lock.
4. Start recording the time with a stop watch at this point.
5. End the time measurement at the stopwatch as soon as the speed which is corresponding to 1100 RPM is read on the onboard/ship's side GPS receiver.
6. Stop recording after reaching the minimum speed.
7. Evaluate the recording after completing the maneuver as described below.



Timing starts with the left cursor position at the start of the deceleration maneuver and ends with the right cursor position at which the speed of the ship is corresponding to 1100 RPM.

This period equates to the deceleration time required by the ship model.

Subsequently compare the two times thus measured (actual time measured with the stop watch and the time required by the ship model).

If the time the ship model needs to attain the speed which is corresponding to 1100 RPM is greater than the time the vessel actually needs to decelerate, the parameter **pTimeConstant_ShipDeceleration** (SysID 1.00x.0050.000804.012.0010.3) must be reduced (ship model decelerates faster).

If the time the ship model needs to attain the speed which is corresponding to 1100 RPM is less than the time the vessel actually needs to decelerate, the parameter **pTimeConstant_ShipDeceleration** (SysID 1.00x.0050.000804.012.0010.3) must be increased (ship model decelerates slower).

3.5 Setting parameters for the “reversing” maneuver

3.5.1 Preparatory steps

To facilitate the procedures described in the sections below, we recommend preparing a monitoring and recording template including the following measuring points:

- ☒ **EngineSpeed** (SysID 1.00x.0010.000501.010.0000.0)
- ☒ **EngineSpedDemandCAN** (SysID 1.00x.0010.000604.013.0000.0)
- ☒ **pIdleSpeedOffset** (SysID 1.00x.001.000518.012.0000.3)
- ☒ **pDisengageWindowHiLimit** (SysID 1..x.0020.043107.154.0000.3)
- ☒ **pShipSpeedDisengage** (SysID 1.00x.0050.000801.014.0020.3)
- ☒ **pEngageWindowHiLimit** (SysID 1.00x.0020.043108.154.0000.3)
- ☒ **ShipSpeed** (SysID 1.002.0050.000801.010.0010.2)
- ☒ **ShipSpeedStatic** (SysID 1.00x.0050.000801.010.0020.2)
- ☒ **pShipSpeedReversing** (SysID 1.00x.0050.000801.014.0030.3)
- ☒ **pNormalEngageShipSpeed** (SysID 1.00x.0050.000801.012.0000.3)
- ☒ **pEngageSpeedOffset** (SysID 1.00x.0080.000517.012.0000.3)
- ☒ **pShipSpeedCrashstop** (SysID 1.00x.0050.000801.014.0040.3)
- ☒ **pCrashEngageShipSpeed** (SysID 1.00x.0050.000801.012.0010.3)
- ☒ **pCrashEngageSpeedOffset** (SysID 1.00x.0080.000517.012.0010.3)
- ☒ **CrashstopActive** (SysID 1.00x.0080.043700.010.0000.2)
- ☒ **Cmd_GearAhead** (SysID 1.00x.0080.049000.013.0030.2)
- ☒ **Cmd_GearNeutral:** (SysID 1.00x.0080.049000.013.0040.2)
- ☒ **Cmd_GearAsternBackflush** (SysID 1.00x.0080.049000.013.0050.2)
- ☒ **InjectionQuantity** (SysID 1.00x.0010.002001.010.0000.0)

3.5.2 Establishing the max coupling limit (pNormalEngageShipSpeed)

Coupling should be possible up to a certain ship speed without any increase in engine speed (“Maneuvering” range, see section 2.3).

The ship speed for the max coupling limit **pNormalEngageShipSpeed** should also be set such that the engine can be coupled reliably without increasing the engine speed.

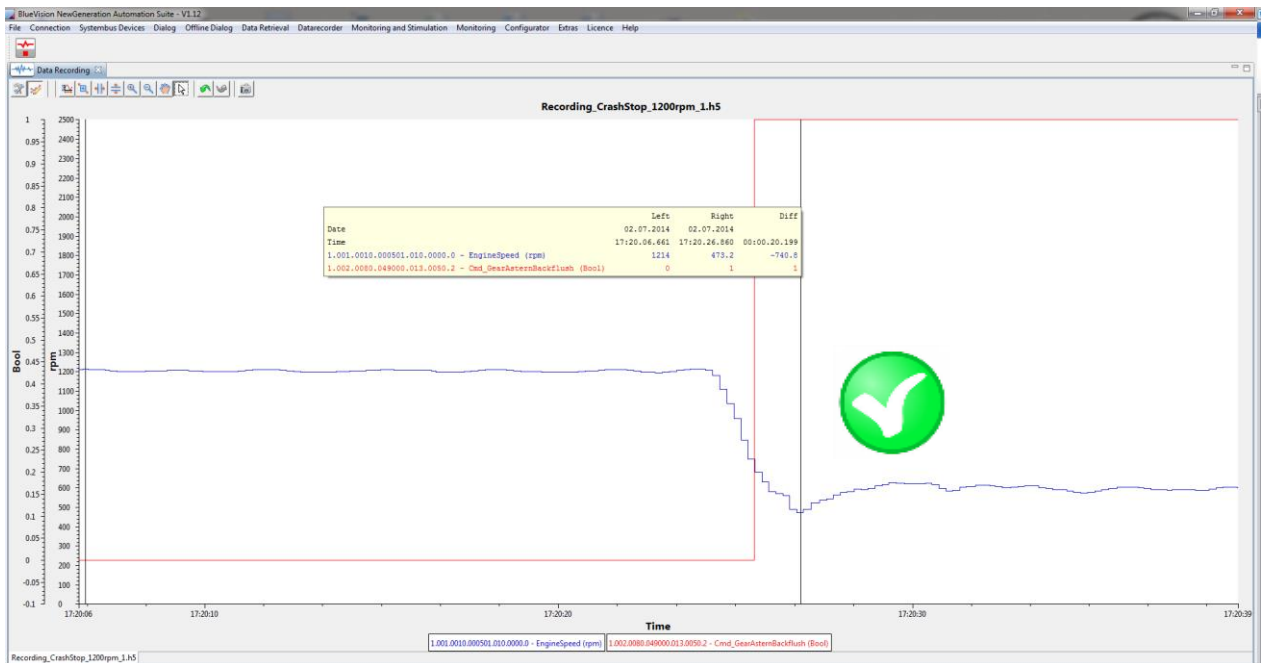
Proceed as follows to establish the ship speed for the maximum coupling limit for the “reversing” coupling maneuver:

1. The parameter for detecting the “crash stop” coupling maneuver **pShipSpeedCrashstop** (SysID 1.00x.0050.000801.014.0040.3) must be temporarily set above the maximum ship speed. This ensures that a “crash stop” coupling maneuver is not detected on performing subsequent steps and prevents the associated parameters taking effect.
2. The parameter for detecting the “reversing” coupling maneuver **pShipSpeedReversing** (SysID 1.00x.0050.000801.014.0030.3) must be temporarily set to the ship speed associated with the engage ahead lock. This ensures that a “reversing” coupling maneuver is detected on performing subsequent steps and allows the associated parameters to take effect.
3. Establish the set value for the maximum coupling limit for the “reversing” maneuver **pNormalEngageShipSpeed** (SysID 1.00x.0050.000801.012.0000.3)
Set this value to 40 % of the max. ship speed at first.
4. Start recording using the template prepared previously or at least with the measuring points **EngineSpeed** and **Cmd_GearAsternBackflush**.
5. Accelerate the ship up to 2 knots above the value set for **pNormalEngageShipSpeed** and wait until the corresponding speed has been reached.
6. Move the propulsion control lever from its current position directly to the engage astern lock.
7. Evaluate the dip in engine speed based on the record.

Ensure that speed dip at the time of coupling does not exceed 25-30% of the idle speed. (Series 2000 600 rpm → 420 rpm, Series 4000 500 rpm → 350 rpm)



Excessive speed dip



Acceptable speed dip

Reduce the maximum coupling limit ***pNormalEngageShipSpeed*** if speed dip is excessive and the engine stalls.

Increase the limit gradually if speed dip is below 20%.

Repeat steps 3 to 7 in this case.

Note:

Having established the max. coupling limit for the “reversing” maneuver, the engine can be relieved a little to increase reliability on coupling by increasing the speed fractionally with ***pEngageSpeedOffset*** (see section 3.5.4).

3.5.3 Detecting the “reversing” coupling maneuver

The “reversing” coupling maneuver is detected by the parameter ***pShipSpeedReversing*** (SysID 1.00x.0050.000801.014.0030.3).

The parameters associated with reversing take effect on detecting a “reversing” coupling maneuver (see parameter overview in Annex).

After establishing the max. coupling limit for the “reversing” maneuver ***pNormalEngageShipSpeed*** (see section 3.5.2), the value of the parameter ***pShipSpeedReversing*** which was temporarily lowered before (detection of the “reversing” coupling maneuver) must be set to the same value as now established for the max. coupling limit ***pNormalEngageShipSpeed***.

3.5.4 Speed offset for coupling maneuver “reversing”

If necessary, the idle speed of the engine may be increased to relieve the engine during the “reversing” coupling maneuver.

This is possible with parameter ***pEngageSpeedOffset*** (SysID 1.00x.0080.000517.012.0000.3).

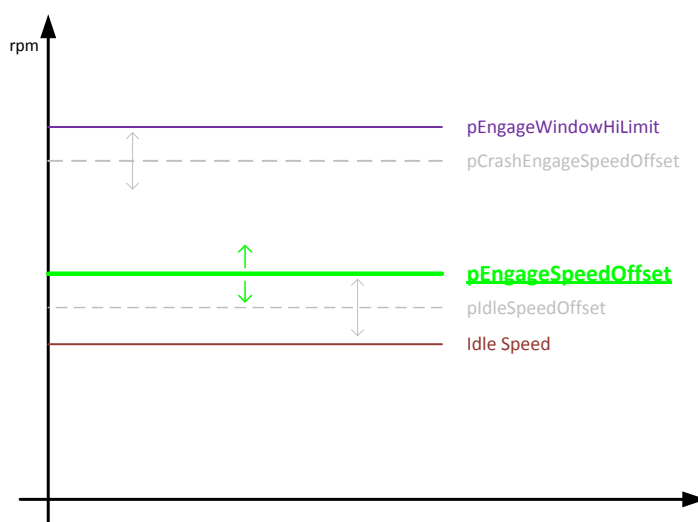
Note:

This speed offset will also take effect during “normal coupling maneuver” in maneuvering.

A speed offset during the coupling sequence exerts a higher degree of mechanical loading on the clutch and gearbox. This leads to increased wear.

This must therefore be discussed with the customer.

The offset described above should not be selected, or should be kept to a minimum, as the next optimization measure (crash stop) involves setting an additional engine speed offset ***pCrashEngageSpeedOffset***.



Note:

Make sure that the engagement speed does not exceed the value set for the maximum admissible engagement window ***pEngageWindowHiLimit***, as engagement is otherwise no longer possible!

3.6 Checking/sequence of the “reversing” coupling maneuver

Perform the following steps to check the “reversing” maneuver:

- Accelerate the ship to a speed above the limit at which a “reversing” coupling maneuver is detected ***pShipSpeedReversing*** (SysID 1.00x.0050.000801.014.0030.3).
- Move the propulsion control lever from its current position directly to the engage astern lock or “full astern”.
- Engine speed / shaft speed is reduced.
- Ship speed is reduced.
- The gearbox is immediately disengaged on reaching the disengagement limit ***pDisengageWindowHiLimit*** (SysID 1.00x.0020.043107.154.0000.3).
- Ship speed continues to decrease.
- Ship speed reaches coupling limit ***pNormalEngageShipSpeed*** (SysID 1.00x.0050.000801.012.0000.3) for engagement astern

AND

the engine speed lies within the engagement window

pEngageWindowHiLimit (SysID 1.00x.0020.043108.154.0000.3)

→ immediate engagement astern.

If one of the two values is not below or within the relevant limit the sequence pauses until this value has been reached. Only then does engagement astern take place.

- The engine does not stall.

3.7 Setting parameters for the “crash stop” maneuver

3.7.1 Establishing the max coupling limit (pCrashEngageShipSpeed)

In order to bring the ship to a standstill faster in the event of an emergency, the parameter **pCrashEngageShipSpeed** allows coupling at a higher ship speed than that set in the **pNormalEngageShipSpeed** parameter.

This requires the ship speed to be above the speed set in the **pShipSpeedCrashstop** parameter (for detection of a “crash stop” coupling maneuver) on initiating the coupling sequence.

A speed offset must be set in addition with the **pCrashEngageSpeedOffset** parameter as the engine would otherwise stall.

Proceed as follows to establish the ship speed for the maximum coupling limit for the “crash stop” coupling maneuver:

1. The parameter for detecting the “crash stop” coupling maneuver **pShipSpeedCrashstop** (SysID 1.00x.0050.000801.014.0040.3) should be temporarily set to the value of the parameter for detecting the “reversing” coupling maneuver **pShipSpeedReversing** (SysID 1.00x.0050.000801.014.0030.3).
2. Define a value x for the speed offset **pCrashEngageSpeedOffset** (SysID 1.00x.0080.000517.012.0010.3).
Select a value at which the idle speed plus the sum of all speed offsets lies just below the admissible engagement window **pEngageWindowHiLimit**.
3. Establish the set value for the maximum coupling limit for the “crash stop” maneuver **pCrashEngageShipSpeed** (SysID 1.00x.0050.000801.012.0010.3)
Set this value to 50 % of the max. ship speed at first.
4. Start recording using the template prepared previously or at least with the measuring points **EngineSpeed** and **Cmd_GearAsternBackflush**.
5. Accelerate the ship up to 2 knots above the value set for **pCrashEngageShipSpeed** and wait until the corresponding speed has been reached.
6. Move the propulsion control lever from its current position directly to the engage astern lock.
7. Evaluate the dip in engine speed based on a record.

Ensure that speed dip on coupling does not exceed 25-30% of the idle speed.
(Series 2000 600 rpm → 420 rpm, Series 4000 500 rpm → 350 rpm)

Reduce the maximum coupling limit **pCrashEngageShipSpeed** if speed dip is excessive and the engine stalls.

Increase the limit gradually if speed dip is below 20%.
Repeat steps 3 to 7 in this case.

3.7.2 Detecting a “crash stop” coupling maneuver

The “crash stop” coupling maneuver is detected by the parameter ***pShipSpeedCrashstop*** (SysID 1.00x.0050.000801.014.0040.3).

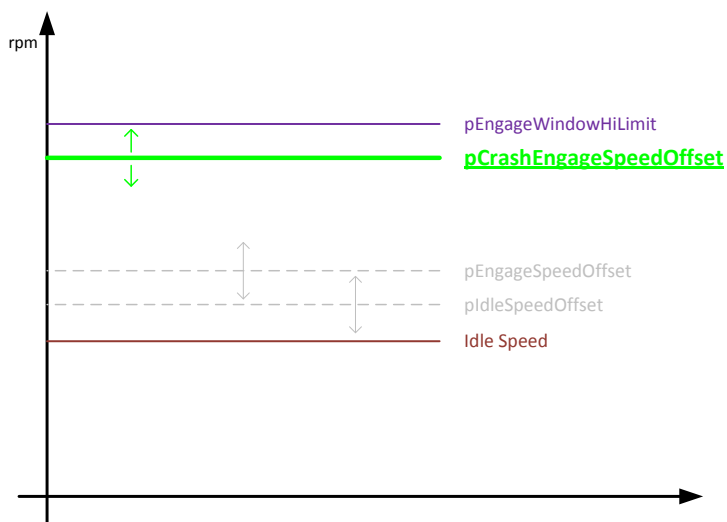
The parameters associated with crash stopping take effect on detecting a “crash stop” coupling maneuver (see parameter overview in Annex).

After establishing the max. coupling limit for the “crash stop” maneuver ***pCrashEngageShipSpeed*** (see section 3.7.1), the value of the parameter ***pShipSpeedCrashstop*** which was temporarily lowered before (detection of the “crash stop” coupling maneuver) must be set to 60-80% of the maximum ship speed. Make sure that this value does not lie below the coupling limit ***pCrashEngageShipSpeed***!

3.7.3 Speed offset for coupling maneuver “crash stop”

Engine idle speed must be increased to relieve the engine during the “crash stop” coupling maneuver.

A “crash stop” coupling maneuver is detected if the ship speed is above the value set for detecting a “crash stop” maneuver ***pShipSpeedCrashstop*** and engine speed is then increased by the ***pIdleSpeedOffset***, ***pEngageSpeedOffset*** and ***pCrashEngageSpeedOffset*** parameters on coupling (see section 3.7.1).



Note:

Make sure that the engagement speed does not exceed the value set for the maximum admissible engagement window ***pEngageWindowHiLimit***, as engagement is otherwise no longer possible!

3.8 Checking/sequence of the “crash stop” coupling maneuver

Perform the following steps to check the “crash stop” maneuver:

- Record the crash stop maneuver.
- Accelerate the ship to maximum speed and wait until the corresponding speed has been reached. This must lie above the limit for detection of a “crash stop” maneuver ***pShipSpeedCrashstop*** (SysID 1.00x.0050.000801.014.0040.3).
- Move the propulsion control lever from the “**full ahead**” position directly to the “**full astern**” setting.
- Engine speed / shaft speed is reduced.
- Ship speed is reduced.
- The gearbox is immediately disengaged on reaching the disengagement limit ***pDisengageWindowHiLimit*** (SysID 1.00x.0020.043107.154.0000.3).
- Ship speed continues to decrease.
- Ship speed reaches coupling limit ***pCrashEngageShipSpeed*** (SysID 1.00x.0050.000801.012.0010.3) for engagement astern

AND

the engine speed lies within the engagement window
pEngageWindowHiLimit (SysID 1.00x.0020.043108.154.0000.3)

→ immediate engagement astern.

If one of the two values is not below or within the relevant limit the sequence pauses until these values have been reached. Only then does engagement astern take place.

- The engine does not stall.
- The ship comes to a standstill.
- The propulsion control lever can be set to the neutral position.

4 Crash Stop Modes

There are 3 different Crashstop modes available.

The chapter 3 of this document describes the Extended Mode setup procedure.

The parameter ***pCrashstop_Mode*** (1.00x.0080.046304.012.0000.3) determines which crash stop behavior is used.

Extended mode (Default) (***pCrashstop_Mode*** = *value 0*)

This mode use separate parameters for

- detection of a reversing (***pShipSpeedReversing***) or a crash stop (***pShipSpeedCrashStop***) maneuver,
- establishing the max. coupling limit for a reversing (***pNormalEngageShipSpeed***) or a crash stop (***pCrashEngageShipSpeed***) maneuver.

Simplified mode (***pCrashstop_Mode*** = *value 1*)

This mode doesn't use separate parameters for the detection and the max. coupling limit.

The detection of a reversing maneuver and the max. coupling limit is just set with parameter ***pShipSpeedReversing***.

The detection of a crash stop maneuver and the max. coupling limit is just set with parameter ***pShipSpeedCrashStop***.

The parameters ***pNormalEngageShipSpeed*** and ***pCrashEngageShipSpeed*** are not used.

The procedure for adjustment is similar to the extended mode.

ShaftSpeed only mode (***pCrashstop_Mode*** = *value 2*)

This mode is only available with shaftspeed sensors!

In this mode the shipmodel is not used!

For engagement, only the shaft speed value is used as a coupling-in criterion.

The shaft speed limit for engagement is set with parameter ***pLM_ShaftSpeedReversing***.

5 Qualification

Whether both the “reversing” and “crash stop” coupling maneuvers are necessary, or if one limit may prove to be enough, depends very much on the vessel concerned and cannot be predicted in advance.

In the case of a planing hull vessel using just one limit may be sufficient as these ships have faster deceleration characteristics by design.

The coupling limit parameters and the parameters for detecting a maneuver could be set to the same value here.

It may even be necessary to deactivate the crash stop feature as it may serve no real purpose in certain applications (depending on ship design and type of service).

This feature may have undesired effects in some circumstances.

A tugboat in which no shaft speed sensors are available would be a good example of such an application. Ship speed may vary significantly when operating at the same engine speed in the case of such vessels. Ship speed is low when towing at high engine speed and higher at the same engine speed when running light. The ship model, however, would always assume the speed to be the same, as it only has one allocation table and is unable to differentiate between variations in use.

When towing, ship speed would be assumed to be higher than is actually the case if the ship model were set with the tugboat running light. The coupling sequence would be unnecessarily delayed.

When running light, ship speed would be assumed to be lower than is actually the case if the ship model were set with the tugboat towing. The coupling sequence would end up stalling the engine as the speed of the boat would be too high.

No clear allocation of engine speed to ship speed is possible here as such boats may run at different speeds depending on their use (running light/towing).

In this case, the crash stop feature must be deactivated with parameter ***pCrashstopDisable*** (SysID 1.00x.0080.043700.012.0000.3).
(1 = deactivated, 0 = activated)

6 Closing remarks

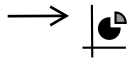
It is simply impossible to describe the many factors, variables and interdependencies affecting a multitude of vessels of varying design in just one document.

It is only possible to recommend certain settings.

The document is intended as a rough guide to better understanding of the functions involved and the general procedure.

7 Annex

7.1 Ship model table (pShipOverShaftSpeed_X0)



7.2 Parameter overview (reversing/crash stop)

